



TRANSFORMER TEST REPORT

DATE:	06/09/2023
ORDER NUMBER:	2023-10003
TRANSFORMER TYPE:	THREE PHASES DRY TYPE TRANSFORMER
SERIAL NUMBER:	D4000240
PROJECT NUMBER:	2022-40405

RATING DATA

RATED FREQUENCY:	50-60	RATED POWER: (KVA)	450
RATED PRIMARY VOLTAGE (V)	400 +/- 2X2,5%	RATED SECONDARY VOLTAGE: (V)	400
RATED PRIMARY CURRENT (A)	650	RATED SECONDARY CURRENT: (A)	650
SHORT CIRCUIT VOLTAGE(Ucc/Un) (%)	6,14	Pj: (W)	5545
CLASS MATERIAL:	F	Po: (W)	1413
CLASS TEMPERATURE:	F	CONNECTION GROUP:	Dyn5
STANDARD IEC:	60076-11		
PROTECTION DEGREE:	IP44	EFFICIENCY %	98,48

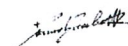
TERMINALS	RESISTANCE (mΩ)
PRI	3,195
SEC	2,712

AMBIENT TEMPERATURE (°C): 25

MEASUREMENT OF VOLTAGE RATIO

V2/V1:	418	/	388,7	=	1,075
	412	/	394	=	1,046
	386	/	420,6	=	0,918

INDUCED OVERVOLTAGE WITHSTAND TEST 2 X Un at 120Hz for 60 sec : RESULT SATISFACTORY


OPERATOR

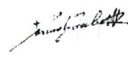
☒ Witnessed	☐ Monitored	☐ Reviewed	
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SHORT CIRCUIT TEST AT 50Hz

PRIMARY			SECONDARY	
TERMINALS	VOLTAGE VALUES (V)	SHORT CIRCUIT CURRENT VALUES (A)	TERMINALS	SHORT CIRCUIT CURRENT VALUES (A)
H1-H2	24,7	652,7	L1-L2	
H2-H3	24,6	654,7	L2-L3	
H1-H3	24,4	645,5		
AVERAGE VAL.:	24,57			
AVERAGE VAL.:		651,0		
COLD Ucc %: 6,1				
COPPER LOSSES (W):	5545			

NO LOAD TEST AT 50Hz

PRIMARY			SECONDARY	
TERMINALS	VOLTAGE VALUES (V)	NO LOAD CURRENT VALUES (A)	TERMINALS	VOLTAGE VALUES (V)
H1-H2 400	401,1 388,7	9,4	L1-L2 400	400 418
H2-H3 400	394	14,8	L2-L3	412
H1-H3 400	420,6	9,3	L1-L3	386
H1 (-5%)	381,0 369,3			
H2 (-5%)	374,3			
H3 (-5%)	399,6			
H1 (+5%)	421,16 408,1			
H2 (+5%)	413,7			
H3 (+5%)	441,63			
AVERAGE VAL.:	401,10			405
AVERAGE VAL.:		11,17		
ACTIVE STEEL LOSSES (W):		1413		


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08 September 2023			
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TEMPERATURE-RISE TEST

TERMINALS	FINAL RESISTANCE (mΩ)	FINAL TEMPERATURE (°C)	TEMPERATURE RISE (K)
	SEE ATTACHED RISE TEST		
STARTING AMBIENT TEMPERATURE (°C):			
FINAL AMBIENT TEMPERATURE (°C):			

LEVEL NOISE TEST AT 1mt

DIRECTION 1 (dB)	DIRECTION 2 (dB)	DIRECTION 3 (dB)	DIRECTION 4 (dB)
52	50	53	51

DIELECTRIC STRENGTH TEST

TERMINALS	A.C. VOLTAGE 50Hz (V)	RESULT
H1-H2-H3 TO EARTH	3000 1MINUTE	PASSED
L1-L2-L3 TO EARTH	3000 1MINUTE	PASSED
H1-H2-H3 TO L1-L2-L3	3000 1MINUTE	PASSED

INSULATION RESISTANCE TEST

TERMINALS	D.C. VOLTAGE (V)	INSULATION RESISTANCE MUST BE OVER 500 MΩ
H1-H2-H3 TO EARTH	1000	>500
L1-L2-L3 TO EARTH	1000	>500
H1-H2-H3 TO L1-L2-L3	1000	>500

WEIGHT CHECK: 1600 Kg

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